AI INDUSTRY INSIGHTS

The ebb and flow of IT people, 2026–2040

A scenario projection of how people move through IT careers: study, a first job, moving up and leaving. For IT job seekers, students and their parents.

STRAVORIS RESEARCH

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The ebb and flow of IT people, 2026-2040

A scenario projection from a system-dynamics model of IT employment: directions and how firm they are, not a forecast.

This note follows people through IT careers in two regions, onshore (clients' home markets in North America and Europe, taken together) and India: who starts studying, who finds a first job, who moves up, who leaves the field, and when each career stage has it hardest. It is written for people looking for IT work, for students weighing an IT degree and for their parents. **Junior, mid and senior** are career stages in the model, not job titles. Every year is compared with 2026, today. The numbers are from the **central case** (the model's calibrated inputs at the central pace of AI adoption) unless stated. Each finding carries a mark: whether its direction holds in every scenario, at both paces of AI adoption and across the range of uncertain inputs, or what it depends on (How firm this is).

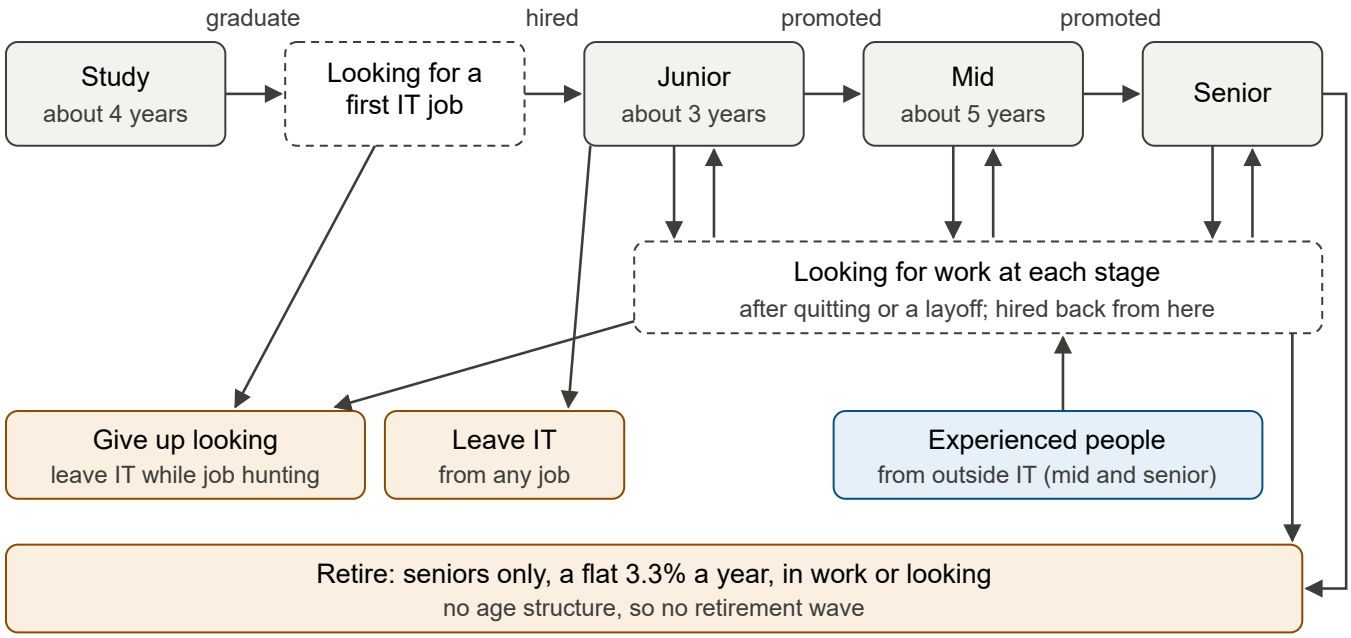
Summary

- **1. Getting a first IT job gets harder, and stays harder.** In the central case, a junior job seeker's chance of being hired within a year averages 21% less onshore and 23% less in India over 2026-2033 than in 2026 (18% and 24% less at the faster adoption pace). Over 2034-2040 it is 22% lower onshore and 35% lower in India, and in 2040 it is still below 2026 in both regions. India's renewed squeeze after 2033 comes from one assumption: in 2040 India has 4% more graduates than in 2026 while junior hiring is 21% lower and junior jobs 35% fewer, because the model lets enrolment grow with demand for IT work (up 51%), not with IT jobs. Where that India result holds in every case, it is because every case shares that assumption.
  Firm in India after 2033. Elsewhere it fails only if AI computing costs jump for a while (to 2033), if AI improves slowly at the faster adoption pace (onshore after 2033), or at the high end of the uncertainty range.
- **2. More graduates for each junior job, and more junior job seekers for each junior in work.** Onshore, graduates per junior hire rise from 1.3 in 2026 to 1.5 over 2026-2033 and 1.6 over 2034-2040 (India: 1.2, 1.4 and 1.6). Job seekers at the junior stage (new graduates still looking for a first job, and juniors between jobs) for each junior in work rise by 61% onshore by 2033 and 68% by 2040 (India 55% and 89%).
  Firm for job seekers per junior in work by 2040. Otherwise it can fail if AI computing costs jump for a while, if AI improves slowly (faster pace), or at the low end of the uncertainty range.
- **3. Mid-level roles take much longer to fill.** Employers take 76% longer to fill a mid-level role onshore and 80% longer in India by 2033 than in 2026; onshore it keeps rising, to 114% longer by 2040. In the central case, 52% of the onshore rise by 2033 and 36% in India come from fewer promotions: there are fewer juniors to promote.
  Firm in every case.
- **4. Fewer mid-level roles are filled by promotion.** Onshore, promotion from junior fills 59% of mid-level roles in 2026, 50% over 2026-2033 and 42% over 2034-2040 (India: 57%, 48% and 42%). Hires from the job market fill more of them. Over 2034-2040, for every mid-level hire from the job market, 0.5 people enter the mid-level job market from outside IT onshore (0.5 in India).
  Firm in every case.
- **5. Students turn away from IT, but the market's growth hides it.** Onshore, enrolment as a share of what it would be with normal job prospects falls from 94% in 2026 to an average of 71% over 2026-2033 and 60% over 2034-2040 (India: 98%, 76% and 69%). The model lets enrolment grow with the IT market, so the number enrolling falls by about a fifth by 2030 (to 79 onshore, 2026 = 100) and is back near its 2026 level by 2040 (98).
  Firm in every case.
- **6. Fewer students choosing IT leaves better odds for those who do.** In a like-for-like test of all 48 pairs (every scenario, both paces, both regions), when students turn away more strongly there are 7-24% fewer graduates over 2026-2040 and junior hiring odds over 2030-2040 are 13-54% higher.
  Firm in every like-for-like pair. Across the uncertain inputs the effect is real but small next to how far AI cuts junior roles.
- **7. IT employment shifts toward senior people.** Juniors' share of IT employment falls onshore from 16% in 2026 to 11% by 2033 and stays there (11% in 2040; India 17% to 12% and 11%). Seniors' share rises from 62% to 70% onshore.
  Firm in every case.
- **8. Senior roles: no steady direction until late in the period.** In the central case, onshore senior roles swing between slower and faster to fill until about 2035, then take longer: 29% longer by 2040, and a senior job seeker's chance of being hired is 13% higher than in 2026. In India senior roles fill faster (26% faster by 2040), which means more competition for senior jobs: the chance of being hired is 22% lower.

Not firm before 2040. By 2040, slower filling onshore holds in 10 of 12 scenarios and across the uncertainty range; faster filling in India holds in all 12, but not at the high end of the uncertainty range (faster pace).

One input drives most of this. How far AI cuts junior roles drives most of the junior and mid-level results: across the uncertain inputs, it alone lines up with juniors' chance of being hired and with how long mid-level roles take to fill with a rank correlation of about 0.97 and 0.97 (1 would be a perfect match). Its value was fitted to the 2024-2026 data, where it is pinned mainly by one fact: junior hiring in 2026 at 65% of its 2024 level (the model gives 65%), from fresher intake at large Indian IT firms and new-graduate hiring in the US. That fixes how far juniors have been cut so far, not how far the cut goes as AI use keeps rising. Its range for the future, from half to one and a half times the fitted value, was set by judgement.

The path through an IT career



Moves between employers happen within each stage and region.

What the stages and regions mean

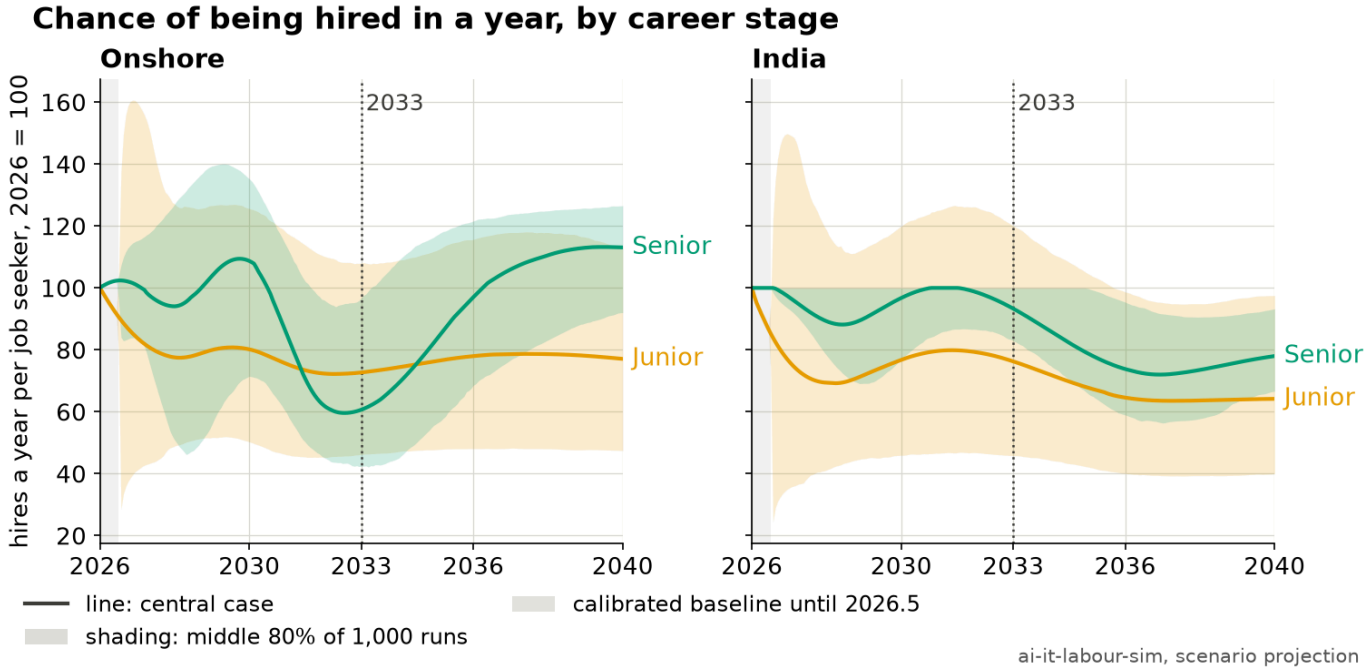
- **Career stages.** In the model a junior is promoted to mid after about 3 years and a mid to senior after about 5 years, when a role is open. A junior's work counts for 0.6 of a mid-level person's and a senior's for 1.4. In 2026, 16% of onshore IT staff are at the junior stage, 22% mid and 62% senior (India 17%, 22% and 60%).
- **Regions.** Each region has its own job market. Onsite staff of Indian IT firms who work onshore on visas are hired from India's job market and count there; when their visa ends they return to it. Some convert to local onshore staff and so move to the onshore job market.
- **Students.** Students study for about 4 years on average and then look for a first job. How many enrol depends on IT unemployment in their region, as students perceive it over about 2 years, and grows with the IT market.

What the measures mean

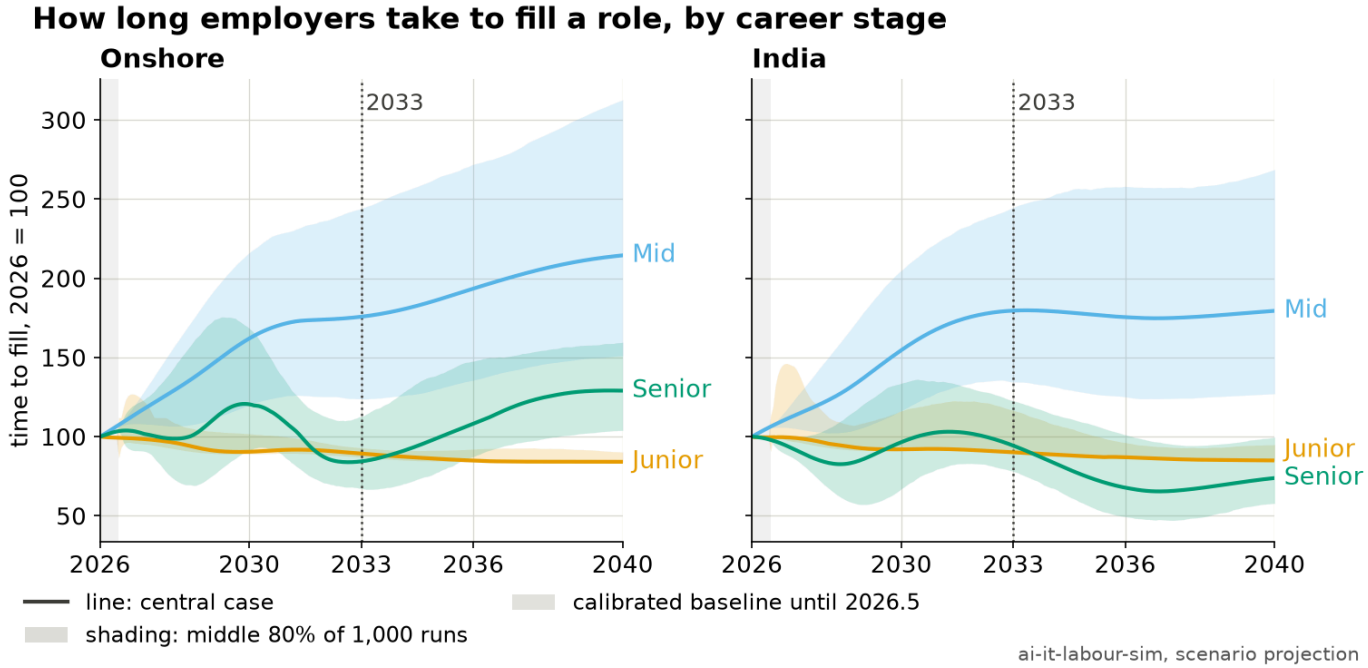
- **Chance of being hired (hiring odds):** hires from the job market in a year for each person looking for work at that stage, in that region. As an index, 2026 = 100, so a value below 100 means lower than in 2026.
- **Job seekers per junior in work:** new graduates still looking for a first job and juniors between jobs, for each junior in work. Shown only against 2026 (2026 = 100): its level is not an unemployment rate and was not calibrated.
- **Time to fill:** how long open roles at a stage take to fill, from promotions, moves between employers and hires from the job market.
- **Graduates per junior hire:** graduates in a year for each junior hired from the job market that year.
- **Flows:** people coming in (graduates, experienced people from outside IT) and leaving (leaving IT from a job, giving up the search, retiring), as a share of everyone in the region's IT job market, in work or looking.

Left out. Freelancers and contractors, nearshore centres, software bought instead of built, business staff doing IT work, and reskilling beyond experienced entry from outside IT. Employer types are combined.

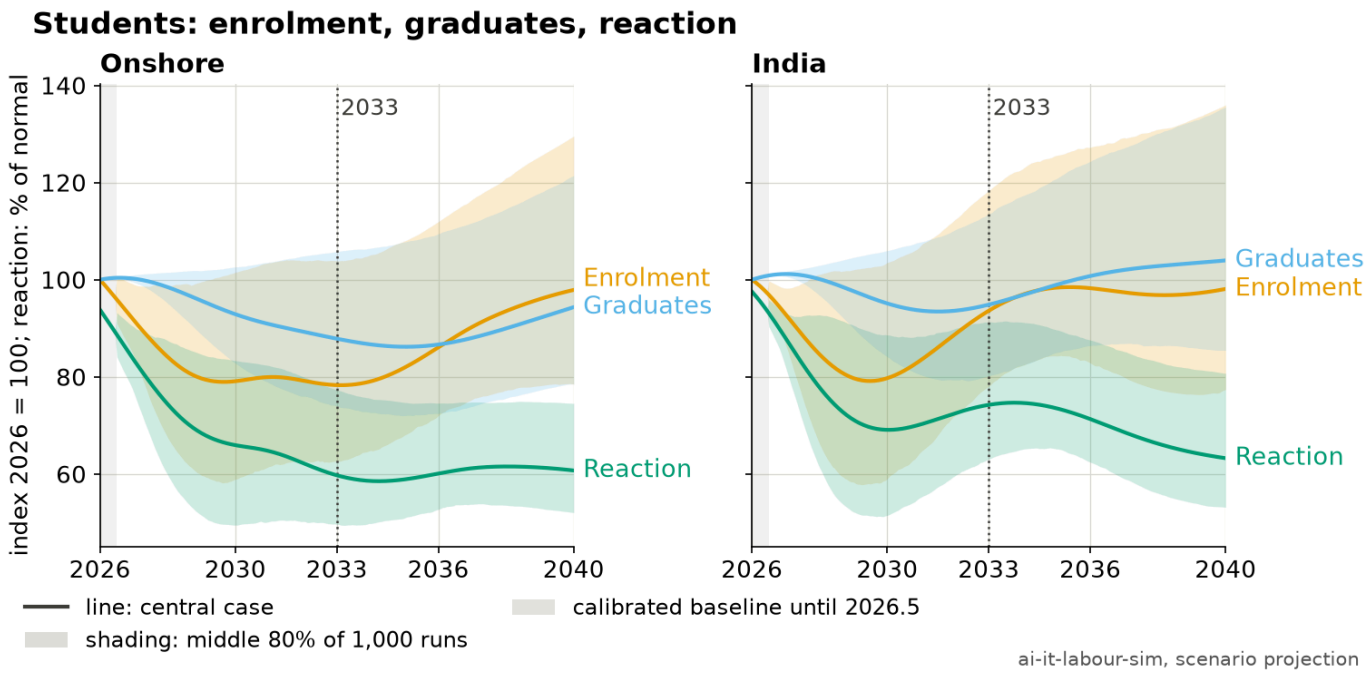
The charts



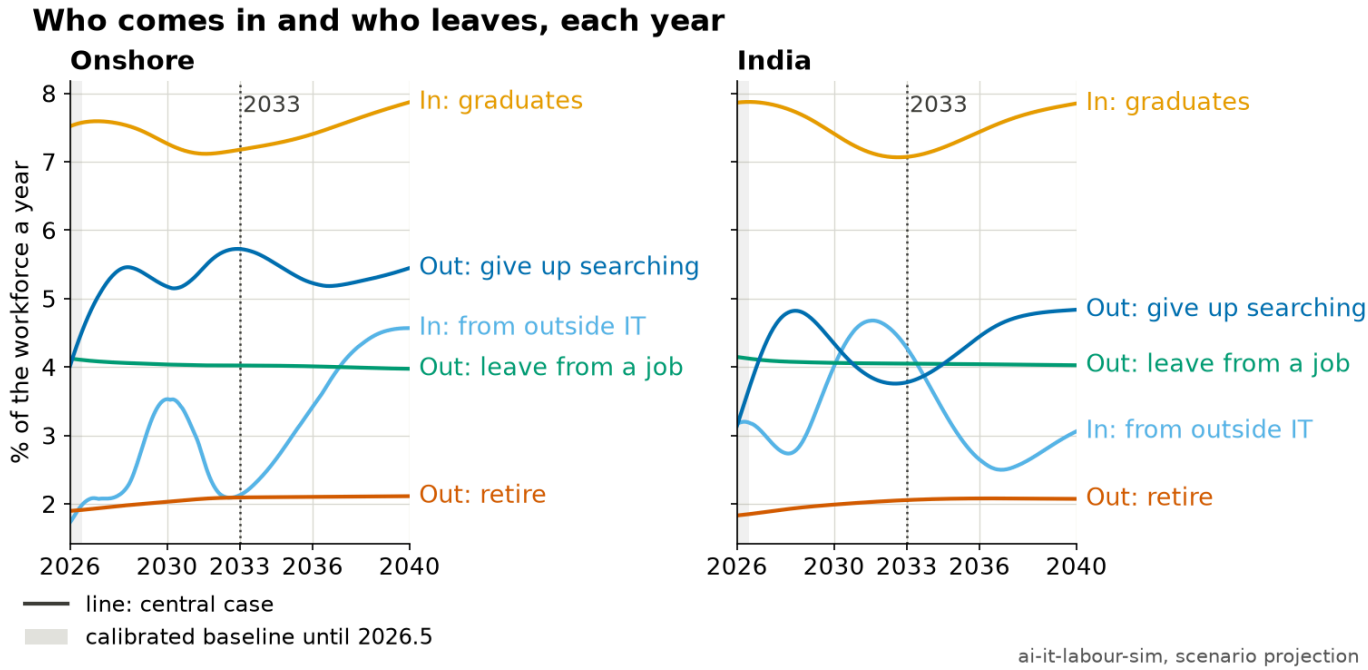
Juniors' chance of being hired falls in both regions; in the central case it averages 21% less onshore over 2026-2033. There is no mid-level line: the model cannot measure how fast a mid-level job seeker is hired (see Assumptions and limits). The shading widens at mid-2026 because one uncertain input, how far AI cuts junior roles, switches on then.



Mid-level roles take longer and longer to fill: 76% longer onshore by 2033 and 114% by 2040. Senior roles take longer to fill onshore in most years, except a few years around 2033; in India they fill faster in most years. Junior roles fill a little faster than in 2026.

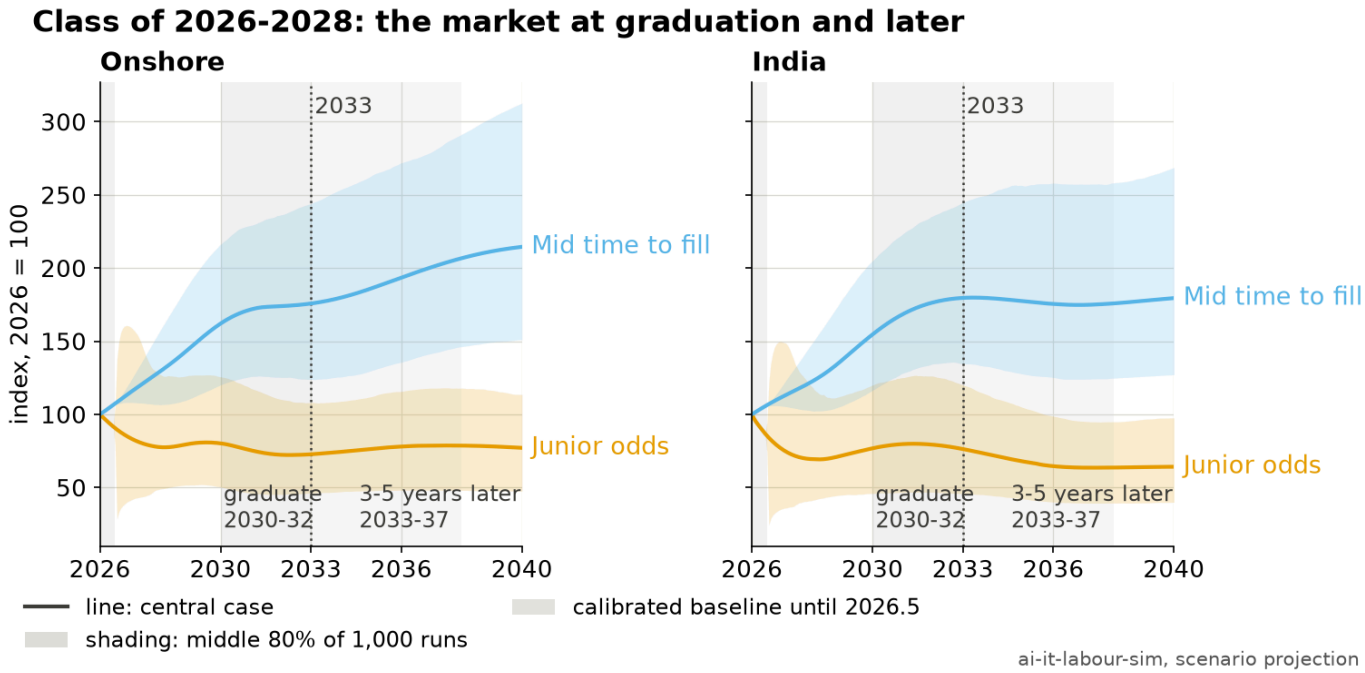


Students' reaction (enrolment as a % of normal) falls to 60% onshore by 2033 and stays low; the number enrolling recovers by 2040 only because the IT market grows. Graduates follow enrolment after a delay.



Each year about 7.4% of the onshore IT job market are new graduates, while 5.2% give up looking and 4.1% leave IT from a job (2026-2033 averages, central case). Experienced people from outside IT come in when roles are hard to fill. No one is laid off in the central case: employers shrink by not replacing people who leave.

For students and parents



Juniors' chance of being hired, and how long mid-level roles take to fill, against 2026. The darker band is when the class of 2026-2028 typically graduates; the lighter one is three to five years later.

The class of 2026, 2027 and 2028

Enrolling in	First year of work	Junior hiring odds, onshore	Junior hiring odds, India	Junior job seekers per junior in work, onshore	Junior job seekers per junior in work, India	Mid time to fill 3-5 years on, onshore	Mid time to fill 3-5 years on, India
2026	2030	78 (48-122)	78 (46-123)	146	151	183	178
2027	2031	74 (46-114)	80 (47-126)	153	149	190	177
2028	2032	72 (46-109)	78 (46-124)	159	152	197	176

All columns are against 2026 (2026 = 100), central case; the uncertainty range is in brackets for hiring odds. Job seekers per junior in work: new graduates still looking for a first job and juniors between jobs, for each junior in work, in the first year of work. First year of work: the calendar year four years after enrolling. Three to five years on: the average of the calendar years three to five years after that. The model cannot measure how fast a mid-level job seeker is hired, so the table shows how long mid-level roles take to fill instead.

What this is, and is not. The model does not follow a class of students. Its students graduate spread around the average of 4 years: about 22% within a year and about 37% after more than four. So these are the market conditions a typical member of each class would meet, not what happens to those people.

Do fewer students mean better odds for those who choose IT? In the model, yes. Running each scenario twice, once with students reacting weakly to scarce jobs and once strongly, the strong reaction gives 7-24% fewer graduates and 13-54% better junior hiring odds over 2030-2040, in every scenario, at both paces and in both regions. Across the uncertain inputs, though, runs with more graduates often have better odds too, because strong demand for IT work raises both; the effect of fewer students is real but smaller than that of AI's cut to junior roles.

Enrolment and graduates. Students react to scarce jobs: enrolment onshore falls to 60% of what normal job prospects would bring by 2033 (India 74%). Graduates follow with a delay: in the central case the number of graduates bottoms out about 2 years after enrolment does in India (enrolment in 2029, graduates in 2031) and onshore (enrolment in 2033, graduates in 2035).

How firm this is

How to read the marks. "Firm in every case" means the direction is the same in all 12 scenarios, at both paces of AI adoption, and across the middle 80% of 1,000 runs with uncertain inputs (the uncertainty range). Otherwise the mark names what the direction depends on. Findings about hiring and flows are tested over the period they describe; findings about levels (time to fill, job seekers per junior in work, the mix of stages) in 2033 and 2040.

Each finding was tested in every scenario, at both paces of AI adoption, and across the uncertainty range. Scenarios: AI improving slowly; AI improving fast; strict regulation; light regulation; a temporary jump in AI computing costs; an IT budget downturn; cheaper IT work drawing in more demand; demand for IT work stalling; faster growth in demand for IT work; clients keeping their appetite for captive centres; clients' appetite for captive centres fading. Findings are numbered as in the summary.

One assumption every case shares. Every scenario and every run lets enrolment grow with demand for IT work, not with IT jobs. India's renewed squeeze on juniors after 2033 comes from it, so where that result holds in every case, it is because every case shares the assumption, not because it is independent of it.

	Test	Tested	Firmness
1	Onshore, 2026-2033	over the period	Holds in 11 of 12 scenarios at both paces; fails if AI computing costs jump for a while; at the high end of the uncertainty range
1	India, 2026-2033	over the period	Holds in 11 of 12 scenarios at both paces; fails if AI computing costs jump for a while; at the high end of the uncertainty range
1	Onshore, 2034-2040	over the period	Holds in 11 of 12 scenarios at both paces; fails if AI improves slowly at the faster adoption pace; at the high end of the uncertainty range
1	India, 2034-2040	over the period	Holds in every case
1	Onshore, still below 2026 in 2040	in the year	Holds in 11 of 12 scenarios at both paces; fails if AI improves slowly at the faster adoption pace; at the high end of the uncertainty range
1	India, still below 2026 in 2040	in the year	Holds in every case
2	Onshore junior seekers per junior in work, 2033	in the year	Holds in 11 of 12 scenarios at both paces; fails if AI computing costs jump for a while
2	India junior seekers per junior in work, 2033	in the year	Holds in 11 of 12 scenarios at both paces; fails if AI computing costs jump for a while; at the low end of the uncertainty range at the faster adoption pace
2	Onshore junior seekers per junior in work, 2040	in the year	Holds in every case
2	India junior seekers per junior in work, 2040	in the year	Holds in every case
2	Onshore graduates per junior hire, 2026-2033	over the period	Holds in 11 of 12 scenarios at both paces; fails if AI computing costs jump for a while at the faster adoption pace; at the low end of the uncertainty range
2	India graduates per junior hire, 2026-2033	over the period	Holds in all 12 scenarios at both paces; fails at the low end of the uncertainty range
2	Onshore graduates per junior hire, 2034-2040	over the period	Holds in 11 of 12 scenarios at both paces; fails if AI improves slowly at the faster adoption pace; at the low end of the uncertainty range
2	India graduates per junior hire, 2034-2040	over the period	Holds in all 12 scenarios at both paces; fails at the low end of the uncertainty range at the faster adoption pace
3	Onshore, 2033	in the year	Holds in every case
3	India, 2033	in the year	Holds in every case
3	Onshore, 2040	in the year	Holds in every case
3	India, 2040	in the year	Holds in every case
4	Onshore, 2026-2033	over the period	Holds in every case
4	India, 2026-2033	over the period	Holds in every case
4	Onshore, 2034-2040	over the period	Holds in every case
4	India, 2034-2040	over the period	Holds in every case
5	Onshore, 2026-2033	over the period	Holds in every case
5	India, 2026-2033	over the period	Holds in every case
5	Onshore, 2034-2040	over the period	Holds in every case
5	India, 2034-2040	over the period	Holds in every case
6	Fewer graduates and better junior odds, like-for-like pairs	2026-2040	Holds in all 48 pairs
7	Onshore, 2033	in the year	Holds in every case
7	India, 2033	in the year	Holds in every case
7	Onshore, 2040	in the year	Holds in every case
7	India, 2040	in the year	Holds in every case
8	Onshore, faster to fill in 2033	in the year	Holds in 9 of 12 scenarios at both paces; fails if AI improves slowly; if AI computing costs jump for a while; if demand for IT work grows faster; at the high end of the uncertainty range

8	Onshore, slower to fill in 2040	in the year	Holds in 10 of 12 scenarios at both paces; fails if demand for IT work stalls; if AI computing costs jump for a while at the central adoption pace
8	India, faster to fill in 2033	in the year	Holds in 4 of 12 scenarios at both paces; fails if AI improves slowly; if AI computing costs jump for a while; if IT budgets go through a downturn; if cheaper IT work draws in more demand; if demand for IT work grows faster; in the central case at the faster adoption pace; if regulation is strict at the faster adoption pace; if clients keep their appetite for captive centres at the faster adoption pace; at the high end of the uncertainty range
8	India, faster to fill in 2040	in the year	Holds in all 12 scenarios at both paces; fails at the high end of the uncertainty range at the faster adoption pace

When things turn

	Central	Faster AI	Scenarios	Uncertainty range
Juniors' hardest year, onshore (lowest hiring odds)	2032	2027	2028-2032	2027-2034
Juniors' hardest year, India	2037	2027	2027-2040	2027-2040
Mid-level time to fill peaks, onshore	2040	2040	2031-2040	2040
Mid-level time to fill peaks, India	2033	2032	2031-2040	2032-2040
Students' reaction bottoms out, onshore	2034	2034	2028-2040	2029-2040
Students' reaction bottoms out, India	2040	2040	2029-2040	2029-2040

The calendar year each turn falls in. 2040 means it has not turned by 2040. Juniors' hardest year is searched from mid-2027, a year after the uncertain inputs switch on, so the switch itself does not set it; 2027 at the start of a range means the low point comes as soon as the search starts. The scenario and uncertainty columns are at the central adoption pace.

What drives the three main findings. Across the uncertain inputs, juniors' hiring odds over 2026-2033 depend most on how far AI cuts junior roles, then how strongly clients choose where to send work by cost and how fast demand for IT work grows after 2026. Mid-level time to fill in 2033 depends most on how far AI cuts junior roles, then how fast checking AI-made work gets cheaper and how fast demand for IT work grows after 2026. Students' reaction depends most on how strongly students turn away from IT when jobs are scarce, then how far AI cuts junior roles and how strongly clients choose where to send work by cost. How far AI cuts junior roles is among the top three for all three, at both paces and in both regions, and first for the first two.

Assumptions and limits

- **Projections, not forecasts.** The results show what the model's structure implies under each scenario. The path to about 2030 rests on data for 2024-2026; later years rest more on structural assumptions. Career delays put much of this story after 2033, where it rests most on those assumptions.
- **Shares and changes only.** The model counts people on its own scale, not in national numbers, so this note gives shares, indices, ratios and percentage changes, never head counts.
- **One onshore region.** Onshore is one generic region of clients' home markets (North America and Europe together in the model's specification), not any one country.
- **Career stages, not job titles, and no ages.** Stages are set by promotion times (about 3 and 5 years) and output. Retirement is a flat 3.3% a year of senior people, so there is no retirement wave from an ageing workforce.
- **AI and juniors.** As AI use rises, employers want fewer juniors: in the central case the junior share of the staff employers want falls from 19% in 2024 to about 9.6% once AI use is complete. AI does not change what juniors can do or how fast they are promoted. If AI let juniors do mid-level work sooner, the mid-level gap would close faster than shown. How far AI cuts junior roles drives most of the uncertainty range; it was fitted to junior hiring up to 2026, and its future range was set by judgement (page 1).
- **Students.** Students react only to IT unemployment in their region, smoothed over about 2 years, and enrolment stays between 40% and 130% of normal; they do not react to news, pay or course costs. Enrolment also grows with demand for IT work, which AI lets grow faster than IT jobs, so the number of students may be overstated late in the period.
- **Classes of students.** The class-of results are the market conditions a typical student meets, not the outcomes of a cohort (page 5).
- **Visas.** Onshore work-visa tightness is held at its calibrated level (0.3 on a scale where 0 is open and 1 is closed) in every year. The visa scenarios are left out.
- **AI adoption pace is the largest unknown.** The 2024-2026 data show the early adoption path but not how fast it continues, so every result is shown at a central and a faster pace.
- **Mid-level job seekers.** The model treats mid-level people as scarce: anyone looking for a mid-level role is hired as fast as the model allows, in every year of every uncertainty run (onshore it slips only briefly, during the temporary jump in AI computing costs). So it cannot show the difficulty many mid-level job seekers face today, and this note makes no finding about it. What it does show is employers' side: mid-level roles taking longer to fill, and more of them filled from outside.

- **Junior job seekers.** New graduates still looking for a first job count as job seekers at the junior stage, against a small number of juniors in work. So the number of junior job seekers for each junior in work is shown only as a change against 2026. Unemployment by stage was not a calibration target: of the measures in this note, only junior hiring and the change in onshore IT unemployment were.
- **No layoffs in the central case.** In the central case employers shrink by not replacing people who leave, not by laying people off. Layoffs appear only in some scenarios: AI improving fast, a temporary jump in AI computing costs, an IT budget downturn, demand for IT work stalling and clients keeping their appetite for captive centres; at the faster adoption pace also, slightly, the central case, strict regulation and light regulation.
- **Other simplifications.** No notice periods; no start-ups (people who leave IT return only as experienced entrants); feedback effects in the model come from outside evidence, not the 2024-2026 data; uncertain input ranges were set by hand.

Method in brief

The model. A system-dynamics model of AI's effect on IT employment for India and onshore, 2024-2040. Client demand for IT work is split between clients' in-house teams, their captive centres in India, global systems integrators, offshore Indian IT firms and boutiques. AI adoption raises the work each person delivers and cuts the share of junior staff employers want. Employers hire to keep up with their work and replace leavers; people move between stages, employers and the job market; students enrol and graduate. It is calibrated to 2024-2026 data and includes feedback effects between employers.

Scenarios and uncertainty. The calibrated path runs unchanged until mid-2026; scenarios and uncertain inputs act from then. 12 scenarios were run: the central case and the eleven named under How firm this is. The uncertainty range comes from 1,000 runs drawing 21 uncertain inputs at once (Latin hypercube sampling, fixed seed), the same draws at both adoption paces. Left out: the two visa scenarios (visa policy is held at its calibrated level), and an extreme stress test, fast AI plus a price shock on offshore Indian IT firms, which is not a plausible future; in it, onshore juniors' hiring odds average 27% below 2026 over 2026-2033.

How the numbers were read. The model's standard outputs give junior hires, time to fill by stage and unemployment by region. Everything else here (hires, promotions and job seekers by stage, enrolment, flows in and out) was read from each step of the model's own calculation and checked against its outputs and its stock-and-flow balance in every run. Timing is read at every step (16 a year). The jump at mid-2026 in some charts is how far AI cuts junior roles switching on in the uncertainty runs.

Version. ai-it-labour-sim 0.2.0, commit `d4ae41de8a569817a7c55a884f3222a60b6a24bf`, tagged `analysis-ready`. Reproduce with `uv run --frozen python temp/analysis/a2-ebb-and-flow/run.py` from the repository root.

Near term: 2027 and 2028

In both years, juniors' chance of being hired is lower than in 2026 in 12 of 12 scenarios at both paces of AI adoption, onshore and in India, but the top of the uncertainty range can be above 2026. Mid-level roles take longer to fill in 2028 than in 2026 in every scenario except the temporary jump in AI computing costs (and, onshore, the IT budget downturn), which hold hiring back for a while. The uncertainty range widens at mid-2026 because one uncertain input, how far AI cuts junior roles, switches on then.

Calendar-year average	Region	2027 central	2027 faster AI	2027 scenarios	2028 central	2028 faster AI	2028 scenarios
Junior hiring odds (2026 = 100)	Onshore	80	74	67-88	78	77	41-95
	India	72	63	61-79	70	64	38-84
Graduates per junior hire	Onshore	1.51	1.61	1.41-1.81	1.52	1.54	1.34-2.64
	India	1.43	1.47	1.33-1.69	1.43	1.42	1.27-2.35
Junior job seekers per junior in work (2026 = 100)	Onshore	129	139	125-135	140	149	126-171
	India	140	155	135-146	153	169	138-197
Mid-level time to fill (2026 = 100)	Onshore	122	119	110-124	137	136	43-148
	India	117	114	109-120	129	128	70-139
Promotion's share of mid-level fills (%)	Onshore	54	54	54-55	51	51	51-55
	India	52	53	52	49	50	49-52
Senior time to fill (2026 = 100)	Onshore	100	92	81-115	102	101	65-150
	India	88	84	72-94	83	81	17-104
Student reaction (% of normal)	Onshore	79	77	78-79	71	67	54-74
	India	83	84	82-83	74	73	64-77
Graduates (2026 = 100)	Onshore	100	99	99-100	97	96	95-98
	India	101	102	100-101	99	99	97-100

Averages over each calendar year; central case (calibrated inputs, central pace of AI adoption), the faster adoption pace, and the range across the scenarios at the central pace.

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